Insurance Claim & Fraud Detection

1. Project Overview

The Insurance Claim & Fraud Detection project focuses on analyzing insurance claims, financial transactions, customer and policy characteristics, and fraud-related indicators using Python, SQL, and Power BI.

The objective is to identify patterns in insurance claims, examine financial inconsistencies, explore characteristics associated with fraud-labelled claims, and present actionable insights through interactive dashboards.

This project demonstrates a complete data analytics workflow, including data preparation, exploratory data analysis, SQL-based validation, financial exception analysis, and interactive data visualization.

2. Problem Statement

Insurance companies process large volumes of claims involving different policy types, claim amounts, incident categories, customer characteristics, and financial outcomes.

Analyzing these records can help identify unusual financial transactions, understand claim patterns, investigate inconsistencies, and improve the visibility of potential risks.

This project addresses the following questions:

- What are the overall patterns in insurance claims?
- How are claims distributed across different statuses, policy types, and incident categories?
- How do claim amounts, approved amounts, and payouts compare?
- Which financial inconsistencies require further investigation?
- What characteristics distinguish claims labelled as fraudulent from those labelled as non-fraudulent?
- How can these findings be communicated through an interactive dashboard?

3. Project Objectives

- Analyze insurance claim data to identify meaningful patterns and trends.
- Clean and prepare the dataset for analysis and visualization.

- Perform data quality and financial consistency checks using SQL.
- Explore fraud-related indicators and compare fraud-labelled and non-fraud-labelled claims.
- Identify claims that violate selected financial validation rules.
- Develop interactive Power BI dashboards to communicate findings.
- Provide data-driven recommendations for further investigation and process improvement.

4. Dataset Description

The project uses an insurance claim fraud dataset containing 100,000 records.

The processed dataset contains 65 columns, including the original dataset fields and three additional financial exception flags.

Important Dataset Fields

Category	Fields
Claim Information	claim_id, claim_date, incident_date, claim_status
Financial Information	claim_amount, approved_amount, payout_amount, deductible
Customer Information	customer_id, customer_name, age, gender, marital_status, occupation, annual_income
Policy Information	policy_id, policy_type, premium_amount, coverage_amount, policy_active
Incident Information	incident_type, accident_severity, incident_location, weather_condition
Vehicle Information	vehicle_make, vehicle_model, vehicle_year, vehicle_value, vehicle_age
Fraud-Related Indicators	fraudulent, fraud_risk_score, suspicious_documents, claim_history_risk
Financial Exception Flags	claim_exceeds_coverage, payout_exceeds_approved, payout_exceeds_coverage

5. Tools and Technologies

Python

Used for data preparation, data cleaning, exploratory data analysis, and dataset processing.

Pandas

Used to manipulate tabular data, inspect missing values, perform transformations, and prepare the processed dataset.

SQL

Used for database setup, data validation, financial consistency checks, claim analysis, fraud analysis, and advanced analytical queries.

Power BI

Used to create interactive dashboards, KPI cards, charts, financial summaries, and analytical visuals.

Git and GitHub

Used to maintain and publish the project source files and documentation.

6. Data Cleaning and Preparation

The dataset was prepared for analysis using Python and Pandas, followed by additional transformations in Power Query.

The preparation workflow included:

- Inspecting dataset dimensions and column names.
- Reviewing data types and missing values.
- Converting relevant columns to appropriate numeric, date, and categorical types.
- Investigating missing values in important fields.
- Preparing the processed dataset for SQL analysis and Power BI reporting.
- Adding financial exception flags to identify potential inconsistencies.

Missing Values

The vehicle_damage column contains 16,527 missing values, representing approximately 16.53% of the dataset.

These missing values were converted from empty strings to null values during data preparation.

Missing values were retained for further consideration rather than automatically assuming that an unknown vehicle damage status meant no damage occurred.

Financial Exception Flags

Three Boolean flags were added to identify claims requiring financial review:

1. claim_exceeds_coverage
2. payout_exceeds_approved
3. payout_exceeds_coverage

These flags allow potentially inconsistent financial records to be isolated and investigated.

7. SQL Analysis

SQL was used to validate the dataset and investigate financial and claim-related patterns.

The project includes the following SQL scripts:

Script	Purpose
01_database_setup.sql	Database setup
02_data_validation.sql	Data quality and validation checks
03_financial_validations.sql	Financial consistency checks
04_claim_analysis.sql	Insurance claim analysis
05_fraud_analysis.sql	Fraud-label and fraud-indicator analysis
06_advanced_sql_analysis.sql	Advanced analytical queries

The SQL analysis supports data validation, aggregation, comparison, exception identification, and investigation of relationships among claim attributes.

8. Exploratory Data Analysis

Exploratory data analysis was conducted to understand the dataset's structure, distributions, missing values, financial measures, and fraud-label patterns.

The analysis included:

- Examining claim and financial amount distributions.
- Comparing fraud-labelled and non-fraud-labelled records.
- Investigating claim statuses and incident characteristics.
- Exploring financial inconsistencies.
- Reviewing the distribution of fraud risk scores.
- Preparing data for dashboard development.

Fraud Label Distribution

Fraud Label	Number of Records	Percentage
No	79,958	79.958%
Yes	20,042	20.042%
Total	100,000	100%

The dataset contains approximately 20.04% fraud-labelled records.

This percentage describes the labels present in the dataset. It should not be interpreted as the actual fraud rate of an insurance company or the accuracy of a fraud detection model.

9. Financial Exception Analysis

Three financial validation rules were applied to identify potentially inconsistent claims.

Financial Validation Rule	Flagged Records
Claim Amount Exceeds Coverage	1,025
Payout Amount Exceeds Approved Amount	19,936
Payout Amount Exceeds Coverage	653

These counts represent records flagged by the respective validation rules.

A claim amount exceeding coverage may require a review of policy limits and claim conditions. A payout exceeding the approved amount may indicate a discrepancy requiring investigation. A payout exceeding coverage may require checking the applicable policy terms and financial records.

The flags identify potential exceptions, not confirmed fraud. Each record requires appropriate review before a conclusion can be reached.

10. Power BI Dashboard

The Power BI report is organized into four pages to present different aspects of the insurance claim dataset without unnecessarily repeating the same analysis.

Page 1: Insurance Claims Overview

Purpose: Provide a high-level summary of insurance claim activity and financial measures.

The page includes KPI cards for:

- Total Claims
- Observed Fraud Rate
- Total Claim Amount
- Total Approved Amount
- Total Payout Amount

Additional charts summarize claim status, average claim amount by incident type, policy type, and average fraud risk score.

Page 2: Fraud Risk Analysis & Detection

Purpose: Explore fraud-labelled claims and fraud-related indicators.

The page includes:

- Fraud Claims
- Observed Fraud Rate
- Non-Fraud Claims
- High-Risk Claims
- Claims by Risk Category
- Fraud-Labelled Claims by Accident Severity

- Fraud-Labelled Claims by Incident Type
- Suspicious Documents by Fraud Label

Date, policy type, and incident type slicers allow users to filter the analysis.

The risk categories are based on the project's defined fraud_risk_score thresholds. They should be treated as exploratory categories rather than independently validated risk classifications.

Page 3: Insurance Claim Financial Analysis

Purpose: Examine claim amounts, financial outcomes, costs, and potential financial inconsistencies.

The page includes KPI cards for:

- Total Deductible
- Average Approved Amount
- Average Payout Amount
- Total Repair Cost

The visualizations include:

- Claimed vs. Approved vs. Paid Amounts
- Claim Amount vs. Payout Amount
- Claim Cost Composition
- Financial Exceptions Requiring Review
- Annual Claim, Approval & Payout Trends (2021–2025)
- Annual Claim Cost Trends (2021–2025)
- A table displaying individual claim-level financial details

The annual trend charts focus on 2021–2025 to avoid presenting the incomplete 2026 period as a full year.

Page 4: Customer, Policy & Operational Insights

Main title: Customer, Policy & Operational Insights

Purpose: Explore customer characteristics, policy information, premium patterns, and claim characteristics.

The planned visuals include:

- Customer-related KPI cards
- Average Policy Premium by Start Year
- Claims by Occupation and Policy Type
- Customer Distribution by Education Level
- Claims by Status
- Claim Amount Drivers Analysis using a decomposition tree

A gauge visual can compare average premium with average coverage as a descriptive indicator, provided its interpretation is clearly explained.

The final page should contain only the visuals that have been implemented and checked in the Power BI report.

11. Key Findings

The following findings are supported by the analysis results shared during project development.

Finding 1: Fraud Label Distribution

The dataset contains 20,042 fraud-labelled claims and 79,958 non-fraud-labelled claims.

The observed fraud-label proportion is approximately 20.04%.

Finding 2: Financial Exceptions

The financial validation rules identified:

- 1,025 claims exceeding coverage.
- 19,936 payouts exceeding approved amounts.
- 653 payouts exceeding coverage.

These records may warrant further review to understand the reasons for the discrepancies.

Finding 3: Missing Vehicle Damage Information

The vehicle_damage column contains 16,527 missing values.

This limits the completeness of analyses that depend on vehicle damage information.

Finding 4: Fraud Risk Score and Label Relationship

The exploratory analysis found that the existing fraud_risk_score separates the supplied fraud labels in the dataset: records labelled non-fraudulent have scores from 0 to 21, while fraud-labelled records have scores from 22 to 97.

This perfect separation raises a concern about potential target leakage or synthetic label construction. Consequently, the risk score should not be treated as independent evidence that a fraud detection method can accurately predict real-world fraud.

12. Business Recommendations

Based on the observed patterns and limitations, the following recommendations are proposed.

1. **Review financial exceptions:** Investigate claims where payouts exceed approved amounts or policy coverage, and determine whether the differences are justified by the applicable policy terms.
2. **Improve data quality:** Investigate missing vehicle damage information and improve the completeness of claim intake records.
3. **Strengthen validation controls:** Introduce appropriate checks to flag inconsistencies among claim amounts, approved amounts, payout amounts, and coverage limits.
4. **Investigate fraud-related indicators:** Use suspicious documents, reporting delays, inconsistent statements, and other available indicators to support structured investigations rather than treating any single indicator as proof of fraud.
5. **Validate fraud labels and risk scores:** Confirm how the existing labels and risk scores were generated before using them to evaluate a predictive fraud detection system.
6. **Monitor claim patterns:** Use the Power BI dashboard to explore changes in claim volume, financial outcomes, customer characteristics, and policy-related patterns.

13. Limitations

Several limitations must be considered when interpreting the results.

- The dataset contains missing values, particularly in the vehicle_damage column.
- Financial exception flags identify records requiring review but do not establish fraudulent activity.
- The existing fraud labels and risk scores show a relationship that may indicate target leakage or synthetic label construction.
- The observed fraud-label percentage may not represent real-world insurance fraud prevalence.
- Descriptive patterns and associations do not establish causal relationships.
- The dataset's findings should not be generalized to all insurance companies without further validation.
- The dashboard provides descriptive and exploratory analysis; it does not independently verify fraud or demonstrate the performance of a trained predictive model.

14. Future Improvements

Potential extensions to the project include:

- Conducting additional data quality checks and documenting their results.
- Investigating the origin and construction of the fraud labels and risk scores.
- Developing a fraud detection model only after verifying the target variable and preventing data leakage.
- Evaluating any predictive model using appropriate validation techniques and metrics.
- Adding more detailed financial exception reporting.
- Improving dashboard interactivity and drill-through capabilities.
- Automating data refresh and report updates.

15. Conclusion

The Insurance Claim & Fraud Detection project demonstrates an end-to-end data analytics workflow using Python, Pandas, SQL, and Power BI.

The project covers data preparation, exploratory analysis, SQL-based validation, financial exception identification, and interactive dashboard development.

The analysis identified financial records requiring review, highlighted missing data, and revealed an important limitation in the relationship between the existing fraud labels and risk scores.

The resulting dashboards organize claim, fraud-label, financial, customer, and policy insights into a structured reporting experience.

Overall, the project demonstrates practical skills in data cleaning, SQL analysis, exploratory data analysis, financial validation, data visualization, and analytical reporting while recognizing the limitations of the available dataset.